

Echo Chain Data Analytics Project Report

1. Project Title

"EchoChain": Circular Economy & Secondary Market Lifecycle Analytics

2. Project Overview

The Echo Chain Data Analytics Project is an end-to-end Business Intelligence solution designed to analyze sales, manufacturing, warranty, and secondary market data. The project integrates Data Visualization technologies to transform raw business data into actionable insights.

The project follows a complete analytics pipeline, beginning with data collection and cleaning, followed by data engineering in Databricks, and finally interactive dashboard development in Power BI.

3. Project Objective

The main objective of this project is to:

- Collect and organize business datasets.
- Perform data cleaning and preprocessing.
- Build a scalable data engineering pipeline using Databricks.
- Store processed data using Delta Lake architecture.
- Create interactive Power BI dashboards.
- Provide executives with meaningful business insights for decision-making.

4. Technologies Used

Technology	Purpose
1.Microsoft Excel	Initial Data Cleaning
2.Visual Studio Code	Project Development
3.Git & GitHub	Version Control
4.Databricks	Data Engineering
5.Delta Lake	Data Storage
6.Power BI	Dashboard Development

Technology	Purpose
7.DAX	KPI Calculations

5. Project Workflow

External Data Collection

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Excel Data Cleaning

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Visual Studio Code + GitHub

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Databricks

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Bronze Layer

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Silver Layer

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Gold Layer



Power BI Integration



Sales Dashboard



Secondary Market Dashboard



Warranty Dashboard



Manufacturing Dashboard



Executive Business Dashboard



KPIs + Charts + Slicers

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Home Page (Bookmark Navigation)

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Project Completed 